Negar Fathi

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Profile

I am a Ph.D. candidate in Computer Science at the University of Nebraska–Lincoln. My research focuses on formal methods, program analysis, and software testing to improve software reliability and correctness. My work also explores how artificial intelligence, including large language models, can be combined with rigorous program reasoning to support software engineering.

Education

Ph.D. in Computer Science *2023–Present*
University of Nebraska–Lincoln, Lincoln, NE, USA
GPA: 3.88/4.00
Research Focus: *Applying formal methods and program analysis techniques to verify safety and liveness properties, with a particular emphasis on termination and non-termination reasoning.*
Advisor: Dr. Rahul Purandare

M.Sc. in Computer Science *2023–2026*
University of Nebraska–Lincoln, Lincoln, NE, USA
GPA: 3.94/4.00
Completed a course-based M.Sc. alongside Ph.D. studies with a focus on advanced computer science coursework.

M.Sc. in Computer Engineering (Software) *2018–2021*
Iran University of Science and Technology, Tehran, Tehran, Iran
GPA: 18.69/20.00 (*Ranked 1st*)
Thesis: *Development of a Constraint Solver to Determine the Domain for Complex Data Types*
Advisor: Dr. Saeed Parsa

B.Sc. in Computer Engineering (Software) *2013–2017*
Babol Noshirvani University of Technology, Babol, Mazandaran, Iran
GPA: 17.29/20.00 (*Ranked 1st*)
Final Project: *Study and Investigation of Routing Protocols in Wireless Sensor Networks*
Advisor: Dr. Mojtaba Mansouri

Skills

Program Verification & Analysis Tools: Roslyn, LLVM/Clang, Z3, KLEE, SeaHorn, Frama-C, DG
Software Testing & Fuzzing: AFL, NUnit, IntelliTest
Programming Languages: C/C++, C#, Python
Frontend Development: HTML, CSS, JavaScript
Backend Development: Entity Framework Core, ASP.NET Core (Web API, MVC)
Databases: Microsoft SQL Server, PostgreSQL
Systems & Development Tools: Docker, Git

Publications

- N. Fathi, H. Unno, T. Terauchi, and R. Purandare, "Sound Termination and Non-termination Analysis of C Programs with Bit-Precise Bounded Semantics and Advanced Constructs," *Proceedings of the ACM on Software Engineering*, vol. 3, no. FSE, pp. 4505–4528, Jun. 2026, doi: 10.1145/3808205.
- A. Kalaee, S. Parsa, and N. Fathi, "COSMOS: A Comprehensive Framework for Automatically Generating Domain-Oriented Test Suite," *Information and Software Technology*, vol. 154, p. 107091, Feb. 2023, doi: 10.1016/j.infsof.2022.107091.
- N. Fathi, H. Unno, T. Terauchi, and R. Purandare, "Loop-Based Slicing and Input-Driven Concretization:

An Empirical Study of Termination and Non-Termination Analysis," *arXiv preprint* arXiv:2607.08988, 2026. [Under review at *IEEE Transactions on Software Engineering*.]
- N. Fathi, "EDFNet: Early Fusion of Edge and Depth for Thin-Obstacle Segmentation in UAV Navigation," *arXiv preprint* arXiv:2604.09694, 2026. [Under review at *Third Workshop on Computer Vision for Geospatial Image Analysis, IEEE/CVF WACV*.]

Experience

Teaching Assistant *Aug. 2025–Present*
University of Nebraska–Lincoln, Lincoln, NE, USA

- Assisted in teaching undergraduate and graduate computer science courses, supported students with course material, graded assignments and exams, and held office hours.
- Courses: CSCE 322 (Programming Language Concepts); CSCE 423/823 (Design and Analysis of Algorithms); CSCE 428/828 (Automata, Computation, and Formal Languages).

Graduate Research Assistant *Aug. 2023–Jul. 2025*
University of Nebraska–Lincoln, Lincoln, NE, USA
Advisor: Dr. Rahul Purandare

- Applied formal methods and program analysis techniques to verify safety and liveness properties, with a particular emphasis on termination and non-termination reasoning.
- Contributed to the NSF-supported project *"Collaborative Research: FMitF: Track II: Enabling Pluggable Runtime Verification Optimizations"* (NSF Award No. 2525242; PI: Dr. Rahul Purandare).

Selected Projects

- **Nexus** Developed a neural-symbolic framework for C/C++ termination and non-termination analysis using LLVM-based loop extraction, LLM-guided witness synthesis, and SMT-based validation.
- **Atropos** Developed a hybrid framework for C/C++ termination and non-termination analysis by combining dynamic loop-entry reachability with static precondition synthesis and refinement.
- **Athena** Developed a sound framework for termination and non-termination analysis of C programs with bit-precise bounded semantics and support for advanced constructs.
- **FocusTNT** Developed a lightweight preprocessing framework for termination and non-termination analysis of C/C++ programs using loop-based slicing and input-driven concretization.
- **llvm2KITTeL** Extended an LLVM-IR-to-LTS translator with arrays, structures, and bit-level operations to produce abstract representations suitable for formal and automated program analysis.
- **COSMOS** Developed a framework for automatically generating domain-oriented test inputs from program path constraints.
- **CleanCode** Developed a Roslyn-based static analysis tool for checking naming, method-structure, and complexity-related clean-code practices in C# programs.
- **SoftModules** Developed a software modularization and architecture recovery tool based on class-level dependency analysis.
- **ModelTests** Developed an object-oriented C# project with interface-based design, stubs, and NUnit-based unit testing.
- **PhysIQ** Developed a hybrid rule-based and machine-learning framework for automated detection and classification of physical-unit inconsistencies in ROS/C++ robotic software.
- **EDFNet** Developed an early-fusion semantic segmentation framework for thin-obstacle detection in UAV navigation using RGB, depth, and edge cues on the DDOS dataset.
- **TaxiClusters** Developed a spatial trajectory analysis project using preprocessing and DBSCAN clustering on large-scale taxi data.
- **WarehouseDB** Developed a warehouse management system integrating PostgreSQL, SQL procedures/functions, and a C# Windows Forms application.
- **SyncAsync** Developed a client-server project comparing synchronous and asynchronous communica-

tion using a WCF service and Windows Forms clients.
- **HospitalFlow** Developed a hospital workflow modeling and improvement project for analyzing current processes and proposing future operational workflows.

Academic Service

- Ad hoc Reviewer, *Machine Intelligence Research*, 2026.

Certifications

- Oregon Programming Languages Summer School (OPLSS): Types, Logic, and Formal Methods *(2025)*; Types, Semantics, and Applications *(2024)*.
- Learning Information Technology Lab (LAITEC), Sharif University of Technology: ASP.NET MVC *(2020)*; ASP.NET *(2019)*; C# Programming *(2019)*.

Honors & Awards

- Othmer Fellowship, University of Nebraska–Lincoln, *2023–2026*
- Ranked 1st among M.Sc. students in Computer Engineering, Iran University of Science and Technology, *2021*
- Ranked 1st among B.Sc. students in Computer Engineering, Babol Noshirvani University of Technology, *2017*

Languages

Persian (Native), English (Fluent)

Last updated: September 28, 2026